1. NAME OF THE MEDICINAL PRODUCT

Beova™ 50mg Film Coated Tablets

2. QUALITATIVE AND QUANTITATIVE COMPOSITION

Each film-coated tablet contains Vibegron 50 mg.

For the full list of excipients, see section 6.1.

3. PHARMACEUTICAL FORM

Film-coated tablet

Light green film-coated tablets with BV50 printed on one side.

4. CLINICAL PARTICULARS

4.1 Therapeutic indications

Treatment of overactive bladder with symptoms of urge urinary incontinence, urinary urgency, and urinary frequency.

4.2 Posology and method of administration

The recommended adult dosage of Vibegron is one 50mg tablet orally, once daily after a meal.

4.3 Contraindications

This product is contraindicated in patients with a history of hypersensitivity to Vibegron or any components of the product.

4.4 Special warnings and precautions for use

<u>PRECAUTIONS CONCERNING INDICATIONS</u>

1 Prior to use of this product, clinical symptoms of patients should be confirmed with an appropriate interview, and diagnosis by exclusion of some other diseases with similar symptoms, including urinary tract infection, urinary calculus, and lower urinary-tract neoplasm such as bladder cancer and prostate cancer, should be made by performing appropriate examinations such as urinalysis. In addition, special examinations should be considered to conduct, if necessary.

2 For patients with lower urinary-tract obstructive disease, including benign prostatic hyperplasia, treatment for these diseases should be prioritised.

<u>PRECAUTIONS CONCERNING PATIENTS WITH SPECIFIC BACKGROUNDS</u>

1 Patients with Complication or History of Diseases, etc.

1.1 Patients with severe cardiac disease

Symptoms may be aggravated due to heart rate increased, etc.

2 Patients with Hepatic Impairment

No dosage adjustment for Vibegron is recommended for patients with mild to moderate hepatic impairment (Child-Pugh A and B). Vibegron has not been studied in patients with severe hepatic impairment (Child-Pugh C) and is not recommended in this patient population.

3 Pediatric Use

No clinical studies in children have been conducted.

4 Geriatric Use

Vibegron should be administered with care while monitoring the condition of the patient. The physiological functions are generally reduced in the elderly. [see 5.2]

5 Urinary Retention

Urinary retention has been reported in patients taking Vibegron. Monitor patients for signs and symptoms of urinary retention, particularly in patients with bladder outlet obstruction and patients taking muscarinic antagonist medications for the treatment of overactive bladder as the risk of urinary retention may be increased. Discontinue Vibegron in patients who develop urinary retention.

6 Patient with renal impairment

No dosage adjustment for Vibegron is recommended for patients with mild (eGFR ≥ 60 - < 90 mL/min/1.73m²), moderate (eGFR ≥ 30 - < 60 mL/min/1.73m²), and severe (< 30 mL/min/1.73m², not on dialysis). Vibegron has not been studied in patient with eGFR < 30 mL/min/1.73m² on dialysis and eGFR < 15 mL/min/1.73m² (with or without hemodialysis) and is not recommended in these patients. [see 5.2]

<u>Precautions for Co-administration (This drug should be administered with caution when co-administered with the following.)</u>

Drugs	Signs, Symptoms, and Treatment	Mechanism and Risk Factors
Azole antifungal drugs Itraconazole, etc HIV protease inhibitors Ritonavir, etc. [See 4.5]	Reportedly, coadministration of Ketoconazole increased blood concentration of Vibegron.	Coadministration of CYP3A4 inhibitors and P-gp inhibitors may increase blood concentration of Vibegron.
Rifampicin Phenytoin Carbamazepine	Pharmacological activity of Vibegron may be reduced.	Coadministration of CYP3A4 inducers and P-gp inducers may decrease blood concentration of Vibegron.
Digoxin	Serum digoxin concentrations should be monitored before initiating and during therapy with Vibegron and used for titration of the digoxin dose to obtain the desired clinical effect. Continue monitoring digoxin concentrations upon	Coadministration with Vibegron may increase maximal concentration (C_{max}) and systemic exposure as assessed by area under the concentration-time curve (AUC).

Drugs	Signs, Symptoms, and Treatment	Mechanism and Risk Factors
	discontinuation of Vibegron and adjust digoxin dose as needed.	

PRECAUTIONS CONCERNING USE

1 Precautions Concerning the Dispensing of the Drug

Patients should be instructed to press the tablet out of a press-through package (PTP) before administration. The sharp corner of swallowed PTP sheet may puncture the esophageal mucosa, resulting in serious complications such as mediastinitis.

4.5 Interaction with other medicinal products and other forms of interaction

Vibegron is suggested to be a substrate for CYP3A4 or P-glycoprotein (P-gp).

Tolterodine

Coadministration of 100 mg [note)] of Vibegron and 4 mg of Tolterodine, a substrate of CYP2D6, to 12 healthy adults resulted in increases in C_{max} and AUC_{0-24} of Vibegron by 1.03-fold and 1.08-fold, respectively, compared with when Vibegron was given alone. Meanwhile, C_{max} and AUC_{0-24} of Tolterodine were increased by 1.12-fold and 1.08-fold, respectively, compared with when Tolterodine was given alone.

Ketoconazole

Coadministration of 100 mg [note)] of Vibegron and 200 mg of Ketoconazole a strong CYP3A4/P-gp inhibitor, to 10 healthy adults resulted in increases in C_{max} and AUC_{inf} of Vibegron by 2.22-fold and 2.08-fold, respectively. [See 4.4]

Diltiazem

Coadministration of 100 mg [note)] of Vibegron and 240 mg of Diltiazem, a moderate CYP3A4/P-gp inhibitor, to 12 healthy adults resulted in increases in C_{max} and AUC_{inf} of Vibegron by 1.68-fold and 1.63-fold, respectively.

Digoxin

Coadministration of Vibegron increased digoxin C_{max} and AUC by 21% and 11%, respectively.

Note) : Approved dose of this drug is 50 mg.

4.6 Fertility, pregnancy and lactation

Pregnant Women

Administration of this product to pregnant women or women who may possibly be pregnant should be allowed only if the expected therapeutic benefits outweigh the possible risks associated with the treatment. Transfer to fetus was reported in animal studies (in rats).

Breast-feeding Women

The continuation or discontinuation of breastfeeding should be considered while taking account of the expected therapeutic benefits and the benefits of maternal feeding. Animal studies (rats) have shown that the drug is excreted into breast milk.

4.7 Effects on ability to drive and use machines

No data available.

4.8 Undesirable effects

The following adverse reactions may occur. Patients should be carefully monitored. If any abnormality is noted appropriate measures should be taken including discontinuation of this drug.

1 Clinically Significant Adverse Reactions

1.1 Urinary retention (incidence unknown)

2 Other Adverse Reactions

	1%≤ to < 2% (Common)	<1% (Uncommon)	Incidence Unknown
Psychoneurological			Headache, Dizziness, Insomnia, Somnolence
Gastrointestinal	Dry mouth, Constipation		Nausea, Abdominal distension, Dyspepsia, Gastritis, Gastrooesophageal reflux disease, Diarrhea, Abdominal pain, Decreased appetite
Cardiovascular			QT prolongation, Palpitations
Urinary/ Renal	Urinary tract infection (Cystitis, etc.), Increased residual urine volume		Urinary hesitation, Bladder pain, Enuresis, Dysuria
Dermatological			Rash, Hyperhidrosis, Pruritus
Ophthalmological		Photophobia	Dry eye, Blurred vision
Hepatic		Increased AST, Increased ALT	Hepatic function abnormal, Increased γ-GTP, Increased Al-P
Others			Fatigue, Hot flush, Hyperlipidemia, Fluid retention, Myalgia, Edema, Increased CK, Thirst, Blood pressure increased, Chest discomfort

4.9 Overdose

There is no experience with inadvertent Vibegron overdosage. In case of suspected overdose, treatment should be symptomatic and supportive.

5. PHARMACOLOGICAL PROPERTIES

5.1 Pharmacodynamic properties

1 Mechanism of Action

Vibegron selectively stimulates β_3-adrenergic receptors present in bladder smooth muscles, and by relaxing bladders, it promotes urine accumulation function, and improves urinary urgency, urinary frequency and urge incontinence in overactive bladder patients.

2 Effects on β-adrenergic receptors

In cells stably expressing the human β_3-adrenergic receptors, Vibegron increased concentrations of intracellular cAMP in a concentration-dependent manner, but in cells expressing human β_1- or β_2-adrenergic receptors it hardly increased concentrations of intracellular cAMP. (*in vitro*)

3 Effects on bladder tissues (*in vitro*)

Vibegron suppressed contraction in a concentration-dependent manner in human bladder tissues contracted by electrical stimulation.

4 Effects on bladder functions

4.1 Vibegron increased the bladder capacity in rhesus monkeys under anesthesia in a dose-dependent manner.

4.2 Vibegron increased the bladder capacity in crab-eating monkeys without anesthesia in a dose-dependent manner.

5.2 Pharmacokinetic properties

<u>Blood Level</u>

1 Single administration

After single oral administration of 10-300 mg note) of Vibegron to 6 healthy adult males under fasted conditions, Cmax and AUCinf rose more than proportional to dose increase, but tmax and t1 / 2 were constant regardless of the dose.

Figure 1. Plasma concentration after single administration of Vibegron under fasted conditions

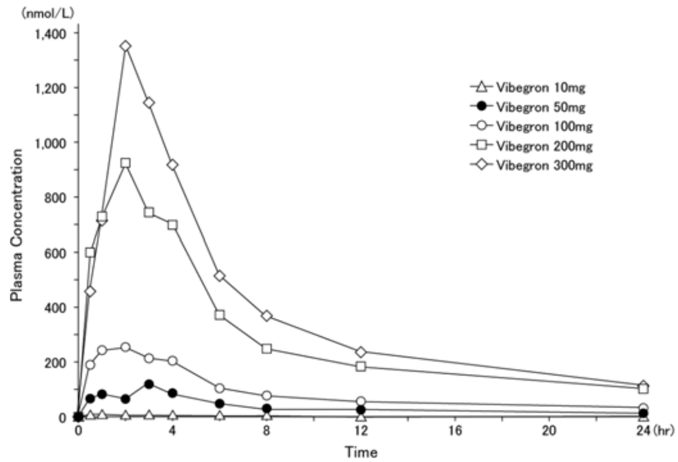

Table 1: Pharmacokinetic parameters after single administration of Vibegron under fasted conditions

Dose	C_{max} (nmol/L)	t_{max} # (hr)	AUC_{inf} (μmol・hr/L)	$t_{1/2}$ (hr)
10mg	6.57 (60.9)	1.00 (1.00-4.00)	0.212 (30.3)	60.5 (40.8)
50mg	134 (34.7)	3.00 (0.800-3.00)	1.92 (27.2)	64.0 (12.6)
100mg	360 (70.3)	2.50 (0.500-4.00)	3.89 (23.1)	58.9 (21.3)
200mg	1090 (40.3)	2.00 (0.500-4.00)	11.5 (16.2)	59.1 (16.7)
300mg	1580 (36.8)	2.00 (1.00-4.00)	13.7 (25.5)	60.7 (15.7)

Geometrical mean (%CV), n = 6, #: median value (minimum value–maximum value)

2 Repeated administration

After repeated oral administration of 50 mg, 100 mg or 200 mg note) of Vibegron once daily for 14 days to 6 healthy adult males under fasted conditions, AUC_{0-24} was ranging from 1.84 to 2.29 times higher than those on Day 1. The steady state of plasma concentrations of Vibegron was achieved by the 7th day of administration.

Table 2: Pharmacokinetic parameters after repeated administration

Dose	Dosing period (Day)	C_{max} (nmol/L)	t_{max} ## (hr)	AUC_{0-24} (μmol•hr/L)	$t_{1/2}$ (hr)
50mg	1	90.1 (73.7)	1.00 (0.500-4.00)	0.559 (69.4)	-
	14#	110 (67.2)	3.00 (0.500-3.00)	1.28 (43.5)	69.6 (9.9)
100mg	1	324 (135.6)	1.50 (0.500-6.00)	1.89 (86.1)	-
	14#	354 (60.3)	2.00 (2.00-4.00)	3.72 (29.6)	64.9 (34.9)
200mg	1	778 (57.4)	2.00 (1.00–4.00)	5.31 (46.3)	-
	14	1380 (28.1)	1.00 (0.500–6.00)	9.76 (14.8)	59.7 (3.3)

Geometrical mean (%CV), n = 6, #: n = 5, ##: median value (minimum value–maximum value)

Absorption

1 Effect of meal

After single oral administration of 50 mg of Vibegron to 8 healthy adult males after a meal, pharmacokinetic parameters were as shown below. C_{max} and AUC_{inf} under fasted conditions were 1.73 and 1.40 times higher, respectively, than those under fed conditions, but there were no effect on T_{max} and $t_{1/2}$.

Table 3: Pharmacokinetic parameters after single administration after a meal

C_{max} (nmol/L)	t_{max} # (hr)	AUC_{inf} (µmol•hr/L)	$t_{1/2}$ (hr)
89.7 (84.3)	1.00 (0.500-2.00)	1.37 (39.7)	68.9 (15.0)

Geometrical mean (%CV), n = 8, #: median value (minimum value–maximum value)

Distribution

The plasma protein binding rate of Vibegron was approximately 49.6%–51.3%. The blood-to-plasma concentration ratio of Vibegron was measured to be 0.8–1.0. *(in-vitro)*

Metabolism

Vibegron mostly remains as unchanged form in human plasma after oral administration, and three different glucuronides and two different oxidative metabolites were identified as metabolites.

Excretion

In the mass balance study of single oral administration of 100 mg [note)] of ^{14}C labeled Vibegron to 6 healthy adult males, 20.3% of the dose was recovered as radioactivity in urine and 59.2% in feces by Day 20. Unchanged form accounted for 92.7% of radioactivity in urine and 91.0% in feces.

Patients with Specific Backgrounds

1 Patients with renal impairment

Among patients with renal impairment, C_{max} and AUC_{inf} after single oral administration of 100 mg [note)] of Vibegron compared with those of healthy adult subjects were 1.96 and 1.49 times higher in mild (eGFR 90–60 mL/min/1.73m^2), 1.68 and 2.06 times higher in moderate (eGFR 60–30 mL/min/1.73m^2), and 1.42 and 1.83 times higher in severe (eGFR less than 30 mL/min/1.73m^2, not on dialysis) renal impairment patients, respectively.

2 Patients with hepatic impairment

After single oral administration of 100 mg [note)] of Vibegron in patients with moderate hepatic impairment (Child-Pugh Classification score of 7-9 points), C_{max} and AUC_{inf} were 1.35 and 1.27 times higher, respectively, than those of healthy adult subjects. [See 4.4]

3 Geriatric use

After repeated oral administration of 100 mg note) of Vibegron once daily for 14 days to healthy elderly males (65 to 74 years, 6 cases), C_{max} and AUC_{0-24} was 1.88 and 1.45 times higher, respectively, than those of healthy non-elderly adult males (23 to 39 years, 5 cases). [See 4.4]

Note) : Approved dose of this drug is 50 mg.

CLINICAL STUDIES

1 Clinical Studies for Efficacy and Safety

1.1 Japan phase III double-blind placebo-controlled study (KRP114V-T301)

Vibegron 50 mg, 100 mg note), or Placebo was orally administered to 1107 patients with overactive bladder once daily after a meal for 12 weeks. Following tables show change in mean urination frequency per day (primary endpoint), change in mean urinary urgency frequency per day and change in mean number of urge incontinence per day (secondary endpoints). Significant improvements were observed in each item evaluated in both treatment groups (Vibegron 50 mg and Vibegron 100 mg) compared with the placebo group.

Adverse reactions occurred in 7.6% (28/370 patients) in the Vibegron 50mg group, 5.4% (20/369 patients) in the Vibegron 100mg group and 5.1% (19/369 patients) in the placebo group. Common adverse reactions were constipation in 1.6% (6/370 patients) and dry mouth in 1.4% (5/370 patients) in the 50 mg group.

Table 1: Change in mean urination frequency per day at Week 12

Treatment group	Cases	At baseline	Change #	Comparison with Placebo ##
Placebo	369	11.20±2.40	-1.21 (-1.40, -1.03)	-
Vibegron 50 mg	370	11.13±2.37	-2.08 (-2.27, -1.89)	p<0.0001
Vibegron 100 mg	368	11.08±2.25	-2.03 (-2.22, -1.84)	p<0.0001

Mean value ± standard deviation
#: Constrained longitudinal data analysis (cLDA), least-squares mean (95% CI)
##: cLDA, a two-tailed significance level of 5% was used

Table 2: Change in mean urinary urgency frequency per day at Week 12

Treatment group	Cases	At baseline	Change #	Comparison with Placebo ##
Placebo	369	3.77±2.23	-1.77 (-1.96, -1.58)	-
Vibegron 50 mg	370	3.70±2.08	-2.28 (-2.46, -2.09)	p=0.0001
Vibegron 100 mg	368	3.77±2.25	-2.44 (-2.63, -2.25)	p<0.0001

Mean value ± standard deviation
#: cLDA, least-squares mean (95% CI)
##: cLDA, a two-tailed significance level of 5% was used. The multiplicity of the test is not adjusted between the evaluation items.

Table 3: Change in mean number of urge incontinence per day at Week 12

Treatment group	Cases	At baseline	Change #	Comparison with Placebo ##
Placebo	333	1.88±1.33	-1.08 (-1.21, -0.96)	-
Vibegron 50 mg	329	1.97±1.48	-1.35 (-1.48, -1.23)	p=0.0015
Vibegron 100 mg	327	1.86±1.31	-1.47 (-1.60, -1.34)	p<0.0001

Mean value ± standard deviation
#: cLDA, least-squares mean (95% CI)
##: cLDA, a two-tailed significance level of 5% was used. The multiplicity of the test is not adjusted between the evaluation items.

1.2 Japan phase III long-term study (KRP114V-T302)

Vibegron at the dose of 50 mg was orally administered to 166 patients with overactive bladder once daily after a meal for 52 weeks. After initial administration of Vibegron at the dose of 50 mg for 8 weeks, dosage was increased from 50 mg to 100 mg note) once daily only in subjects who met the following: insufficient in efficacy, investigator judged no problem in safety, and the subject wished to increase dosage. Following tables show changes at Week 8 and 52 in mean urination frequency per day, mean urinary urgency frequency per day, and mean number of urge incontinence per day. In both the Vibegron 50 mg dose-maintained group and the Vibegron 100 mg dose-increased group, improvement from the baseline (before administration) was observed in every endpoint, which was maintained without attenuation over the duration of 52 weeks.

Adverse reactions occurred in 18.1% (21/116 patients) in the Vibegron 50 mg dose-maintained group, and 11.8% (6/51 patients) in the Vibegron 100 mg dose-increased group. Common adverse reactions were increased residual urine volume in 4.3% (5/116 patients), dry mouth and cystitis both in 2.6% (3/116 patients), and constipation in 1.7% (2/116 patient) in the Vibegron 50 mg dose-maintained group, and constipation and dry mouth both in 3.9% (2/51 patient), and rheumatoid arthritis and pruritus both in 2.0% (1/51 patient) in the Vibegron 100 mg dose-increased group.

Table 4: Change in mean urination frequency per day

Treatment group	Cases	At baseline	Change at Week 8 #	Change at Week 52 #
Dose-maintained group Vibegron 50 mg	115	10.62±1.88	-2.52 (-2.90, -2.13)	-2.71 (-3.21, -2.21)
Dose-increased group Vibegron 100 mg	51	12.78±2.95	-1.85 (-2.42, -1.28)	-3.16 (-3.90, -2.41)

Mean value ± standard deviation
#: Longitudinal data analysis (LDA), least-squares mean (95% CI)

Table 5: Change in mean urinary urgency frequency per day

Treatment group	Cases	At baseline	Change at Week 8 #	Change at Week 52 #
Dose-maintained group Vibegron 50 mg	115	3.79±2.59	-2.72 (-3.14, -2.29)	-2.91 (-3.43, -2.39)
Dose-increased group Vibegron 100 mg	51	5.56±3.57	-2.19 (-2.82, -1.55)	-3.42 (-4.20, -2.65)

Mean value ± standard deviation
#: LDA, least-squares mean (95% CI)

Table 6: Change in mean number of urge incontinence per day

Treatment group	Cases	At baseline	Change at Week 8 #	Change at Week 52 #
Dose-maintained group Vibegron 50 mg	93	1.81±1.39	-1.49 (-1.78, -1.20)	-1.55 (-1.88, -1.22)
Dose-increased group Vibegron 100 mg	45	2.79±2.60	-1.49 (-1.91, -1.07)	-2.29 (-2.76, -1.81)

Mean value ± standard deviation
#: LDA, least-squares mean (95% CI)

Note) : Approved dose of this drug is 50 mg.

5.3 Preclinical safety data

Carcinogenesis, Mutagenesis, Impairment of Fertility

Carcinogenesis

No carcinogenicity was observed in long-term studies conducted in mice and rats treated with daily oral doses of Vibegron for approximately 2 years. In the mouse carcinogenicity study, CD-1 mice were treated with daily oral doses of Vibegron up to 90 mg/kg/day in males and up to 150 mg/kg/day in females, corresponding to estimated systemic exposures (AUC) 21-and 55-fold higher, respectively, than in humans treated with the recommended daily dose of Vibegron. In the rat carcinogenicity study, Sprague Dawley rats were treated with daily oral doses of Vibegron up to 30 mg/kg/day in males and up to 180 mg/kg/day in females, corresponding to systemic exposures (AUC) 18-and 117-fold higher, respectively, than in humans treated with the recommended daily dose of Vibegron.

Mutagenesis

Vibegron was not mutagenic in in vitro microbial reverse mutation assays, showed no evidence of genotoxic activity in an in vitro human peripheral blood lymphocyte chromosomal aberration assay, and did not increase the frequency of micronucleated polychromatic erythrocytes in an in vivo rat bone marrow micronucleus assay.

Impairment of Fertility

In fertility/general reproductive toxicity studies conducted in rats, females were treated with daily oral doses of 0, 30, 100, 300, or 1000 mg/kg/day Vibegron and males were treated with daily oral doses of 0, 10, 30, or 300 mg/kg/day Vibegron. No effects on fertility were observed in female or male rats at doses up to 300 mg/kg/day, associated with systemic exposure (AUC) at least 274-fold higher than in humans treated with the recommended daily dose of Vibegron. General toxicity, decreased fecundity, and decreased fertility were observed in female rats at 1000 mg/kg/day, associated with estimated systemic exposure 1867-fold higher than in humans treated with the recommended daily dose of Vibegron.

6. PHARMACEUTICAL PARTICULARS

6.1 List of excipients

D-Mannitol, Microcrystalline cellulose, Croscarmellose sodium, Hydroxypropylcellulose, Magnesium stearate, Lactose hydrate, Hypromellose, Titanium oxide, FD&C Blue No.2 Aluminium Lake, Triacetin, Yellow ferric oxide, Carnauba wax

6.2 Incompatibilities

Not applicable

6.3 Special precautions for storage

Do not store above 30ºC.

6.4 Shelf life

Vibegron should be used before the expiration date indicated in the product carton.

6.5 Nature and contents of container

PVC/Aluminum blister packs of 28 tablets [14 tablets (PTP) × 2].

7. MANUFACTURER

Manufactured by:
KYORIN Pharmaceutical Group Facilities Co., Ltd. Noshiro Plant.
1, Matsubara, Noshiro-shi, Akita 016-0000, Japan

Repacked by:
Bora Pharmaceuticals Co., Ltd.
No.54, Gong-Yeh W. Road, Guan-Tyan District, Tainan City, Taiwan

8. PRODUCT REGISTRATION HOLDER

Eisai (Malaysia) Sdn Bhd
Unit 701D, Level 7, Tower D, Uptown 5,
No. 5, Jalan SS21/39, Damansara Uptown,
47400 Petaling Jaya, Selangor, Malaysia

9. DATE OF REVISION

November 2024